

Problem

Three equal impedances, $60 + j30 \, \Omega$ each, are delta-connected to a 230-V rms, three-phase circuit. Another three equal impedances, $40 + j10 \, \Omega$ each, are wye-connected across the same circuit at the same points. Determine:

(a) the line current

(b) the total complex power supplied to the two loads

(c) the power factor of the two loads combined

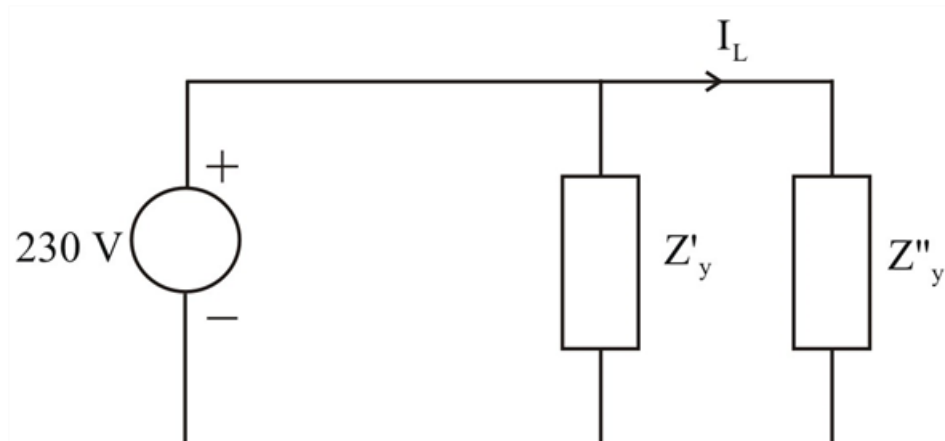
Step-by-step solution

Step 1 of 5

(A) This is a balanced three phase system and we can use per – phase equivalent circuit. The delta connected load is converted to its WYE – connected equivalent.

$$Z_y^* = \frac{1}{3} Z_0 = \frac{(60 + j30)}{3} = 20 + j10$$

Step 2 of 5



Step 3 of 5

$$Z_y = Z'_y \parallel Z_y^* = (40 + j10) \parallel (20 + j10) = 13.5 + j5.5$$

$$I_L = \frac{230}{13.5 + j5.5} = 14.61 - j5.953 \text{ A}$$

Step 4 of 5

$$(B) \bar{S} = V_s I_L^* = 3.361 + j1.368 \text{ KVA}$$

Step 5 of 5

$$(c) \quad Pf = \frac{P}{S} = 0.9261 \quad .$$